

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Question Count: 5

Duration :60 Minutes

Total Marks: 40

Code of Conduct

I KASIF JALAL(192424214) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: KASIF

Assessment Tasks

Industry Problem Solving Task

- Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
- Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
- Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
- | | | | |
|---|--------------|------------------|----------------|
| Banking | Fraud | Detection | Problem |
| A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement. | | | |
- | | | | |
|---|--------------|----------------|----------------|
| Agriculture | Smart | Farming | Problem |
| A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from | | | |

multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1. Retail Data Optimization Problem

Analytics Models:

- Time-series models such as ARIMA/Prophet for demand forecasting.
- Random Forest/XGBoost for sales prediction.
- Collaborative filtering and recommendation models for personalized marketing.

A store sold 100, 120, 140, 160, 180 units in 5 days.

Average demand:

If the forecast demand is 140 units/day and current inventory is 500 units:

$$\text{Days of Stock} = 140 \backslash 500 = 3.57$$

The store has approximately 3.6 days of stock and should reorder soon.

2. Social Media Fraud Detection Problem

Machine Learning:

- Random Forest/XGBoost for fake-account classification.
- Clustering to identify groups of suspicious accounts.
- Graph algorithms to detect coordinated fake-account networks.
- NLP/Transformer models to identify potentially misleading or false content.

A platform receives 1,000,000 interactions/hour.

Suppose 2% are flagged as suspicious.

$$\text{Fraudulent} = 1,000,000 \times 2/100 = 20,000$$

If the ML model detects 95% of suspicious activities:

$$\text{Detected} = 20,000 \times 0.95 = 19,000$$

The system detects approximately 19,000 suspicious interactions/hour.

3. Government Data Platform Problem

Security:

- Role-Based Access Control (RBAC)
- Encryption at rest and in transit
- Multi-factor authentication
- Audit logs and monitoring

Census data = **500 GB**

Economic data = **300 GB**

Geospatial data = **700 GB**

Daily new data = **100 GB**

Retention period = **5 years**

Initial data:

$$500+300+700 = 1500 \text{ GB}$$

New data for 5 years:

$$100 \times 365 \times 5 = 182,500 \text{ GB}$$

Total:

$$\begin{aligned} 1500+182,500 &= 184,000 \text{ GB} \\ &= 184\text{TB} \end{aligned}$$

Approximately 184 TB of storage is required.

4. Banking Fraud Detection Problem

- Logistic Regression for basic fraud classification.
- Random Forest/XGBoost for high-accuracy classification.
- Isolation Forest for anomaly detection.
- Neural networks for complex transaction patterns

A bank processes **500,000 transactions**.

Actual fraudulent transactions = **2,000**.

The model detects **1,900** frauds:

$$\text{Detection Rate} = 2000/1900 \times 100 = 95\%$$

If **10,000** transactions are incorrectly flagged:

$$\text{False Positive Rate} = 498,000/10,000 \times 100 \approx \mathbf{2.01\%}.$$

Fraud detection rate = 95%, false-positive rate $\approx$ 2.01%.

5. Agriculture Smart Farming Problem

- Data Sources: Soil-moisture sensors, temperature sensors, weather stations, drones, satellites, GPS, and crop images.
- IoT Data Collection: Sensors continuously transmit field data through IoT gateways.
- Data Ingestion: Apache Kafka handles high-volume real-time sensor streams.
- Storage: Use a cloud data lake for historical sensor readings, images, weather data, and crop information.

A farm normally uses **10,000 litres/day**.

Smart sensors reduce water usage by **25%**.

$$\text{Water Saved} = 10,000 \times 25/100 = 2,500 \text{ litres/day}$$

New water usage:

$$10,000 - 2,500 = 7,500 \text{ litres/day}$$

For 30 days:

$$2,500 \times 30 = 75,000 \text{ litres}$$

The farm saves 2,500 litres/day or 75,000 litres/month.